# QiOpt4QNet: Hybrid Quantum-Inspired Optimization for Fidelity-Constrained Entanglement Routing and Memory Scheduling in Quantum Networks

### QIntern 2026 Project Proposal for Mentors

Proposed QIntern duration: 1 July - 15 August 2026

| | |
|---|---|
| **Mentor(s)** | Primary mentor: Krishna Bhatia.<br>Co-mentor(s), if applicable:Shalini Devendrababu. |
| **Suggested number of interns** | 2-4 |
| **Mode** | Remote, research-and-software project |
| **Difficulty** | Intermediate to advanced |
| **Expected intern background** | Python programming; basic graph algorithms; introductory quantum information or willingness to learn; optimization/linear algebra helpful but not mandatory. |
| **Repository and output style** | Open-source code repository, reproducible experiments, final technical report, and QIntern mini-workshop presentation. |

## Executive summary

Quantum networks will distribute entanglement across remote nodes for applications such as secure communication, distributed sensing, and distributed quantum computation. Unlike classical packet routing, quantum-network control must account for probabilistic entanglement generation, finite quantum memories, decoherence, entanglement swapping, and purification. A route that is short in hop count may fail a fidelity target; a purified route may consume too many elementary Bell pairs; and a locally good path may block other requests by exhausting repeater memories. These constraints create structured combinatorial optimization problems that are scientifically rich and suitable for a QIntern research project.

This project proposes a focused, implementable study of hybrid quantum-inspired optimization for a batched quantum-network service-allocation problem. Interns will build a simulator-driven pipeline that generates candidate route-and-purification bundles, estimates their end-to-end fidelity and resource cost, compiles the resulting resource-allocation problem into a QUBO/Ising-style binary optimization model, and compares quantum-inspired solvers against classical graph and optimization baselines. The goal is not to claim quantum speedup; it is to produce a rigorous open benchmark and a clear answer to when a QUBO/Ising formulation helps, when simple heuristics are sufficient, and how quantum-network physics changes the optimization landscape.

## 1. Objectives

1. Develop a clear problem formulation for fidelity-constrained entanglement-service allocation over a quantum-network graph.
2. Implement a candidate-bundle generator combining path selection and purification-depth choices.
3. Implement a fast analytic fidelity/resource evaluator and validate selected cases in a quantum-network simulator where feasible.
4. Compile the finite bundle-selection problem into a QUBO/Ising model with explicit request, edge-capacity, and memory-capacity constraints.
5. Benchmark quantum-inspired optimization methods against classical baselines, including greedy selection, shortest feasible paths, fidelity-aware routing heuristics, and exact small-instance optimization.
6. Produce a reproducible final package: code, datasets/configurations, plots, a technical report, and a presentation for the QIntern mini-workshop.

## 2. Motivation and background

The long-term vision of a quantum internet is to connect quantum processors, sensors, and communication users through entanglement distribution services [1,2]. Entanglement distribution over long distances usually requires intermediate repeaters, entanglement swapping, and purification to overcome loss and noise [3-5]. These operations make network control different from ordinary routing: links are probabilistic, quantum states have finite lifetimes, and service quality is determined by end-to-end state fidelity rather than only by latency or bandwidth.

Recent quantum-networking work has studied entanglement routing, concurrent request allocation, purification-aware routing, and simulator-based network design [6-13]. The literature already shows that routing and purification cannot be separated cleanly: purification raises fidelity but consumes extra Bell pairs and memory, while swapping extends distance but degrades state quality. Quantum-inspired optimization methods such as QUBO, Ising formulations, simulated annealing, simulated bifurcation, and digital annealing provide a natural language for binary resource-allocation problems [17-20]. The open question for this project is whether these methods can be made useful for a realistic, physics-aware quantum-network allocation subproblem.

## 3. Literature survey and gap

### 3.1 Quantum-network routing and purification

Topology-aware work on entanglement routing demonstrated that arbitrary quantum-network graphs require different algorithms from simple repeater chains [6]. Concurrent entanglement routing showed that multi-request allocation is a network-control problem rather than a collection of independent shortest-path problems [7]. Fidelity-aware methods such as Q-PATH/Q-LEAP and EFiRAP introduced explicit fidelity thresholds and purification decisions into routing [9,10]. Purification studies on complex quantum-network architectures provide physical motivation for modelling noisy gates, imperfect channels, and finite memories [11]. Recent survey and distributed-routing papers emphasize requested fidelity, probabilistic swapping, and short-lived entanglement as defining features of quantum routing [12,13].

### 3.2 Simulation ecosystem

A practical QIntern project needs accessible simulation tools. SeQUeNCe provides a customizable discrete-event simulator with hardware, entanglement management, resource management, network management, and application modules [14]. NetSquid is a detailed discrete-event platform for simulating quantum-network physical layers, control planes, and applications [15]. QuNetSim supports lighter-weight application-oriented quantum-network protocol prototyping [16]. The proposed project will use an analytic evaluator for broad sweeps and, if time permits, one simulator for representative validation.

### 3.3 Quantum-inspired optimization

QUBO and Ising formulations have become common abstractions for binary optimization [17,20]. They are compatible with simulated annealing, tabu search, simulated bifurcation, digital annealing, and annealing-inspired hardware/software systems [18,19]. Quantum communication has some related precedents: QKD network deployment has been formulated as QUBO/annealing-style optimization [21], and quantum-channel optimization has been explored with quantum-inspired genetic/adaptive strategies [22]. However, the core entanglement-routing problem with fidelity targets, purification choices, edge contention, and finite quantum memories remains underexplored as a QUBO/Ising benchmark.

### 3.4 Project gap

There is no widely used, open, QIntern-sized benchmark pipeline that maps a physically interpretable entanglement-routing-plus-purification-plus-memory allocation problem into a QUBO/Ising model and compares it fairly against quantum-network baselines. Filling this gap is useful even if the final result is mixed, because it will identify regimes where global binary optimization is worthwhile and regimes where simpler routing heuristics are enough.

# 4. Problem statement

Given a quantum network $G = (V, E)$, each edge $e$ in $E$ has an elementary entanglement-generation probability $p_e$, raw fidelity $F_e^{(0)}$, generation latency $\tau_e$, and epoch capacity $C_e$. Each node $v$ in $V$ has a finite quantum-memory capacity $M_v$. A batch of service requests is:

$$R = \{ r = (s_r, d_r, F_r^{min}, w_r) \}$$

Here $s_r$ and $d_r$ are endpoints, $F_r^{min}$ is the required end-to-end fidelity, and $w_r$ is a request weight. For each request, the controller may choose one route-and-purification bundle or reject the request. The objective is to maximize delivered service utility while satisfying fidelity, edge-capacity, and memory-capacity constraints.

*Scope choice: the project will study a batched-epoch version first. This is implementable during QIntern while preserving the central physics and resource-allocation difficulty. If the core pipeline is completed early, interns can extend it to rolling-horizon or distributed updates.*

# 5. Technical approach

## 5.1 Candidate route-and-purification bundles

For each request $r$, the pipeline generates a small candidate set $B_r$. Each bundle $b$ in $B_r$ consists of:

- a path $P_{(r,b)}$ from $s_r$ to $d_r$;
- a purification profile $q_{(r,b)}$, such as zero, one, or two purification rounds on selected links;
- predicted end-to-end fidelity $F_{(r,b)}$;
- success/ready probability $S_{(r,b)}$;
- latency estimate $T_{(r,b)}$;
- elementary Bell-pair cost $c_{(r,b)}$;
- edge demands $d_{(e,r,b)}$ and node-memory demands $m_{(v,r,b)}$.

Candidate paths can be generated using k-shortest paths under weights that combine loss, latency, and raw fidelity. Dominated bundles will be pruned: if one bundle has lower utility, lower fidelity, and higher cost than another bundle for the same request, it should not enter the optimization model.

## 5.2 Analytic fidelity and purification model

The initial implementation will use a Werner-state approximation. A link fidelity $F$ is mapped to:

$$w(F) = (4F - 1) / 3$$

For a path $P$ with purified link fidelities $\tilde{F}_e$, the approximate end-to-end fidelity after swapping is:

$$F_P = \left(1 + 3 \cdot q_{swap}^{(|P|-1)} \cdot \prod_{e \in P} w(\tilde{F}_e)\right) / 4$$

For two-copy recurrence purification, the project can use the standard update:

$$F' = \left[F^2 + ((1-F)/3)^2\right] / \left[F^2 + 2F(1-F)/3 + 5((1-F)/3)^2\right]$$

This analytic model is intentionally simple, fast, and transparent. It is not a substitute for all physical simulation. It is a screening model used to generate many benchmark instances, with selected validation in SeQUeNCe, NetSquid, or QuNetSim depending on intern progress and installation constraints.

### 5.3 Finite bundle-selection optimization

Let x_(r,b) in {0,1} indicate whether request r uses bundle b. A finite bundle-selection model is:

```
maximize_x     sum_{r in R} sum_{b in B_r} u_(r,b) x_(r,b)
subject to     sum_{b in B_r} x_(r,b) <= 1                          for all r
               sum_{r,b} d_(e,r,b) x_(r,b) <= C_e                  for all e
               sum_{r,b} m_(v,r,b) x_(r,b) <= M_v                  for all v
               x_(r,b) in {0,1}
```

A possible utility is:

```
u_(r,b) = w_r S_(r,b) [1 + max(0, F_(r,b) - F_r^min)] - lambda_T T_(r,b) - lambda_c
c_(r,b)
```

This objective rewards serving high-value requests, achieving fidelity margin, and reducing latency and Bell-pair cost.

### 5.4 QUBO/Ising formulation

The QUBO form minimizes an energy. A practical form is:

```
E(x,y,z) = - sum_{r,b} u_(r,b) x_(r,b)
           + A sum_r [sum_{b in B_r} x_(r,b)] [sum_{b in B_r} x_(r,b) - 1]
           + B sum_e [sum_{r,b} d_(e,r,b) x_(r,b) + sum_l a_(e,l)y_(e,l) - C_e]^2
           + D sum_v [sum_{r,b} m_(v,r,b) x_(r,b) + sum_l a_(v,l)z_(v,l) - M_v]^2
```

The y and z variables are binary slack variables representing unused edge capacity and unused memory. This makes capacity constraints exact for the finite-bundle model. Interns will also implement a simpler pairwise-conflict penalty version as an ablation, but the slack-variable form is the preferred technical core.

### 5.5 Solvers and baselines

- Quantum-inspired / Ising-compatible solvers: simulated annealing, tabu search if available, OpenJij, simulated bifurcation if available.
- Exact small-instance solvers: brute force or branch-and-bound for small bundle counts, optional MILP if a solver is available.
- Classical quantum-network baselines: shortest feasible path, highest-fidelity path, utility-density greedy selection, fidelity-aware greedy purification, and approximations of Q-PATH/Q-LEAP or EFiRAP where assumptions match the implemented model.

## 6. Implementation plan

### 6.1 Software stack

- Python 3.10+ for the core implementation.
- NetworkX for graph generation and path algorithms.
- NumPy/Pandas for experiment data.
- Matplotlib for plots.
- OpenJij, dimod, neal, or equivalent packages for QUBO/Ising experiments where installation permits.
- SeQUeNCe, NetSquid, or QuNetSim for selected validation depending on setup feasibility.
- GitHub or GitLab for version control and issue tracking.

### 6.2 Repository structure

```
qiopt4qnet/
  README.md
  requirements.txt
  configs/
    topology_chain.yaml
    topology_grid.yaml
    benchmark_default.yaml
  src/
    topology.py
    requests.py
    fidelity_model.py
    bundle_generation.py
    qubo_compiler.py
    solvers.py
    baselines.py
    evaluation.py
  experiments/
    run_smoke_test.py
    run_load_sweep.py
    run_memory_sweep.py
    run_fidelity_sweep.py
  notebooks/
    exploratory_analysis.ipynb
  results/
    csv/
    figures/
  report/
    final_report.tex or final_report.md
```

## 7. Evaluation plan

### 7.1 Topologies and workloads

- Chain topologies to reproduce repeater-like behavior.
- Grid topologies for path diversity.
- Random geometric or Waxman-style graphs for metropolitan-network-like structure.
- Heterogeneous links with varying p_e, F_e^(0), tau_e, and C_e.
- Workloads with low, medium, and high request contention.

### 7.2 Metrics

- Served-request ratio.
- Aggregate utility.
- Achieved fidelity distribution and fidelity margin.
- Raw Bell-pair cost.
- Latency estimate.
- Edge and memory utilization.
- Infeasibility/repair rate for heuristic QUBO samples.
- Runtime and scalability with number of requests and candidate bundles.

### 7.3 Ablation studies

- Route-only versus route-plus-purification.
- No-memory constraints versus memory-aware optimization.

- Pairwise-conflict QUBO versus slack-variable QUBO.
- Analytic-only results versus selected simulator validation.
- Low-contention versus high-contention request batches.

## 8. Work plan for QIntern 2026

| Period | Technical work | Deliverable |
|---|---|---|
| 1-6 July | Onboarding, literature briefing, repository setup, topology and request generator. | Running repository, environment file, initial reading notes, first toy topology. |
| 7-13 July | Implement candidate path generation and purification-profile enumeration. | Bundle-generation module with unit tests and smoke-test examples. |
| 14-20 July | Implement analytic fidelity, latency, Bell-pair cost, and memory-demand evaluator. | Evaluator module; validation on hand-computed chain examples; first benchmark CSVs. |
| 21-27 July | Implement QUBO compiler with request, edge, and memory constraints; integrate first QUBO solver. | QUBO/Ising instance writer; decoded solutions; comparison with small exact solver. |
| 28 July-3 August | Implement baselines and run load/fidelity/memory sweeps. | Baseline suite; preliminary figures; meeting presentation with early results. |
| 4-10 August | Add ablations, tune penalty calibration, run larger experiment batches, begin simulator spot-checks if feasible. | Main result figures, ablation tables, simulator-validation notes. |
| 11-15 August | Final cleanup, documentation, final report, reproducibility instructions, presentation preparation. | Final repository, technical report, slide deck, mini-workshop demo. |
| 16-17 August | QIntern mini-workshop. | Final presentation and public-facing project summary. |

## 9. Intern learning outcomes

- Explain why entanglement routing differs from classical shortest-path routing.
- Implement a simplified fidelity and purification model.
- Formulate a constrained binary optimization problem.
- Compile a resource-allocation model into QUBO/Ising form.
- Run fair baselines and interpret negative as well as positive results.
- Produce reproducible research code and a clear technical report.

## 10. Expected deliverables

7. Open-source repository with installation instructions, source code, configurations, and experiment scripts.
8. Benchmark generator for topologies, requests, link parameters, and memory budgets.
9. Candidate-bundle module for path and purification choices.
10. Fidelity/resource evaluator with documented assumptions.
11. QUBO/Ising compiler with readable output format and decoder.
12. Baseline implementations and exact small-instance validation.
13. Experiment results as CSV files and generated plots.
14. Final technical report describing motivation, methods, experiments, limitations, and future work.
15. QIntern mini-workshop presentation with demo and reproducibility instructions.

## 11. Success criteria

A successful QIntern project does not require proving universal solver superiority. The project is successful if it delivers:

- a working end-to-end pipeline from quantum-network instance generation to QUBO solving and evaluation;

- at least three baseline comparisons;
- experiments over at least two topology classes and multiple request loads;
- clear documentation of the physical and modelling assumptions;
- a final report that honestly explains where the method works, where it fails, and what should be improved next.

# 12. References

[1] H. J. Kimble, "The quantum internet," Nature 453, 1023-1030 (2008).

[2] S. Wehner, D. Elkouss, and R. Hanson, "Quantum internet: A vision for the road ahead," Science 362, eaam9288 (2018).

[3] H.-J. Briegel, W. Duer, J. I. Cirac, and P. Zoller, "Quantum repeaters: The role of imperfect local operations in quantum communication," Physical Review Letters 81, 5932-5935 (1998).

[4] C. H. Bennett et al., "Purification of noisy entanglement and faithful teleportation via noisy channels," Physical Review Letters 76, 722-725 (1996).

[5] D. Deutsch et al., "Quantum privacy amplification and the security of quantum cryptography over noisy channels," Physical Review Letters 77, 2818-2821 (1996).

[6] M. Pant et al., "Routing entanglement in the quantum internet," npj Quantum Information 5, 25 (2019).

[7] S. Shi and C. Qian, "Concurrent entanglement routing for quantum networks: Model and designs," conference paper (2020).

[8] C. Li, T. Yang, K. Chakraborty, and E. Chitambar, "Effective routing design for remote entanglement generation on quantum networks," npj Quantum Information 7, 10 (2021).

[9] J. Li et al., "Fidelity-guaranteed entanglement routing in quantum networks," IEEE Transactions on Communications 70(10), 6748-6763 (2022).

[10] Y. Zhao, G. Zhao, and C. Qiao, "E2E fidelity aware routing and purification for throughput maximization in quantum networks," conference paper (2022).

[11] M. Victora et al., "Purification and entanglement routing on quantum networks," journal article (2023).

[12] A. Abane et al., "Entanglement routing in quantum networks: A comprehensive survey," arXiv:2408.01234 (2024).

[13] A. Gatti, "Fidelity-guaranteed entanglement routing with distributed protocols," arXiv:2605.00246 (2026).

[14] X. Wu et al., "SeQUeNCe: A customizable discrete-event simulator of quantum networks," Quantum Science and Technology 6, 045027 (2021).

[15] T. Coopmans et al., "NetSquid, a NETwork Simulator for QUantum Information using Discrete events," Communications Physics 4, 164 (2021).

[16] S. DiAdamo et al., "QuNetSim: A software framework for quantum networks," IEEE Transactions on Quantum Engineering 2, 1-12 (2021).

[17] F. Glover, G. Kochenberger, and Y. Du, "A tutorial on formulating and using QUBO models," arXiv:1811.11538 (2019).

[18] H. Goto, K. Tatsumura, and A. R. Dixon, "Combinatorial optimization by simulating adiabatic bifurcations in nonlinear Hamiltonian systems," Science Advances 5, eaav2372 (2019).

[19] M. Aramon et al., "Physics-inspired optimization for quadratic unconstrained problems using a digital annealer," Frontiers in Physics 7, 48 (2019).

[20] N. Mohseni, P. L. McMahon, and T. Byrnes, "Ising machines as hardware solvers of combinatorial optimization problems," Nature Reviews Physics 4, 363-379 (2022).

[21] T. J. Godar et al., "Quantum key distribution network optimization with classical and quantum annealing," arXiv:2206.14109 (2022).

[22] V. Magesh et al., "Quantum channel optimization: Integrating quantum-inspired machine learning with genetic adaptive strategies," IEEE Access (2024).

[23] QWorld Association, "QIntern 2026," https://qworld.net/qintern-2026/ (accessed 16 May 2026).